

Neurology Life Hacks

Rewiring your brain

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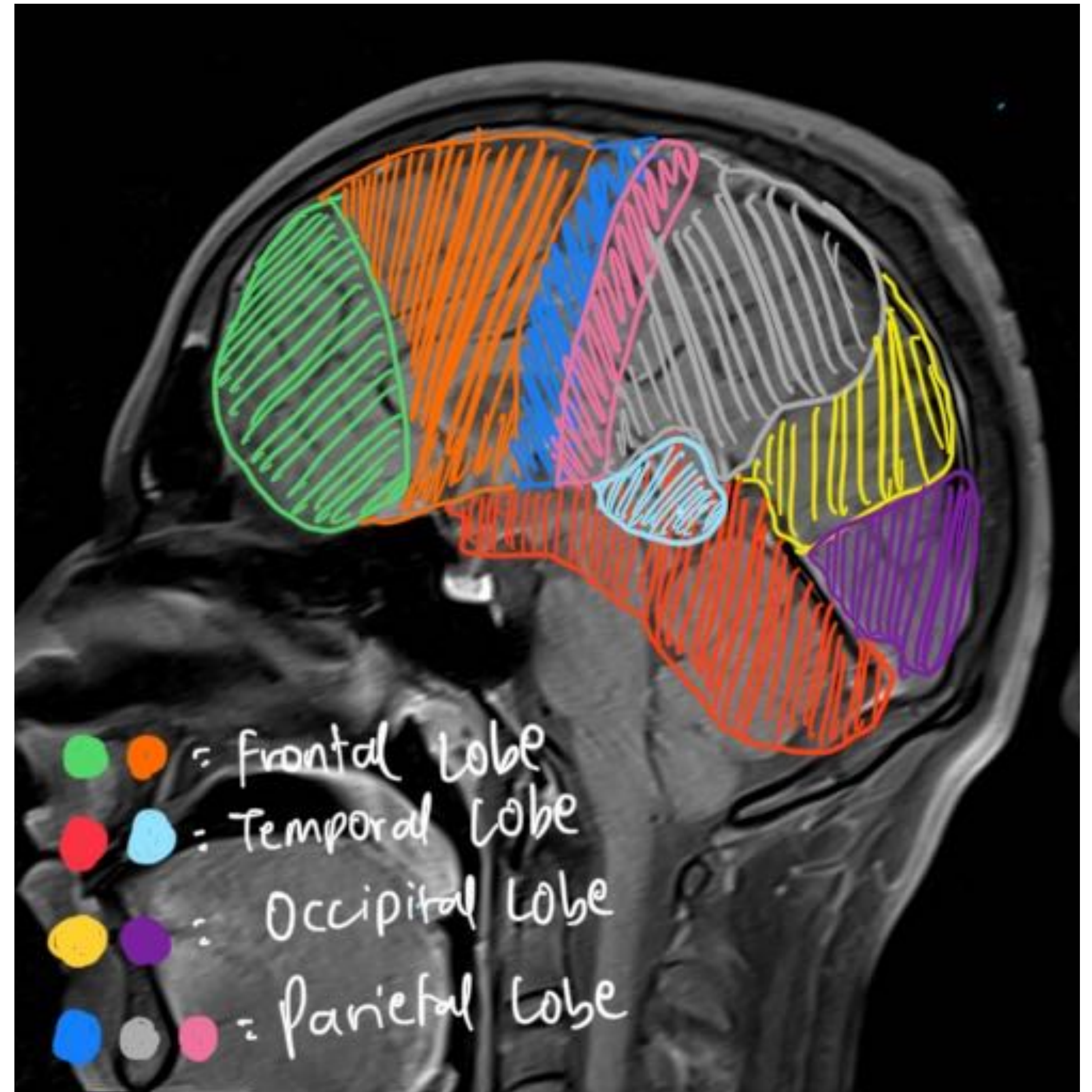
The OG

- This is what our (my) brain looks like in general.
- Fits well inside the cranium
- This is how living beings think, move, react and feel.
- Super complex system of neurons + the brain + spinal cord make up the entire CNS



Lobes and Cortexes

- Every section of the brain plays its own part and develops at varying rates
- Frontal Lobe = absorbs sensory information (light, sound, touch) and takes action
- Temporal Lobe = Auditory perception, understanding semantics for both speech and vision
- Occipital Lobe = Visual perception
- Parietal Lobe = Spatial sense, navigation and touch feedback



Frontal Lobe

- Primary Motor Cortex: Contralateral movement
- Broca area: Producing speech
- Frontal eye fields: Moving the eyes
- Prefrontal cortex (my fave): Restraint, Initiative, order
- How to train it:
 1. Basically doing hard things. This will buff up your anterior mid-cingulate cortex.
 2. Avoid information overload
 3. Memory training games

Occipital Lobe

- Visual / visual association cortex: receiving and recognizing visual stimuli. Determining colors, identifying faces etc
- How to train it:
 1. Mahjong actually is pretty cool way to do it
 2. Learn braille / ASL
 3. Team sports like Soccer

Parietal Lobe

- Primary somatosensory cortex:
Movement etc
- This is where the usual hand eye coordination happens
- How to train it:
 1. Juggling
 2. Dance
 3. Mental 3D simulations
 4. Balancing activities

Temporal Lobe

- Primary auditory cortex:
Everything to do with sounds
- How to train it:
 1. Try to make music
 2. Maybe sing
 3. Learn a new language
 4. Rhythm based games

Catching some ZZZZZ's

How sleep plays a huge role



Why its important

- While we are asleep, most of our basic functions are suspended while certain specialized functions become active.
- We put our brain and body through the gauntlet for about 16/24 hours, so we need those 8 hours for maintenance and upkeep.
- This is how our brain improves its ability to learn, and the body to fight infections, repair muscles etc.
- Getting a good night's sleep entails:
 1. Get enough sunlight during civilian hours
 2. Avoid naps about 5 hours before scheduled bed time
 3. Avoid workouts within 1-2 hours of hitting the sack
 4. A warm bath will change ur life

Substances that modulate sleep

Its always just melatonin vs caffeine. Use wisely.



Caffeine

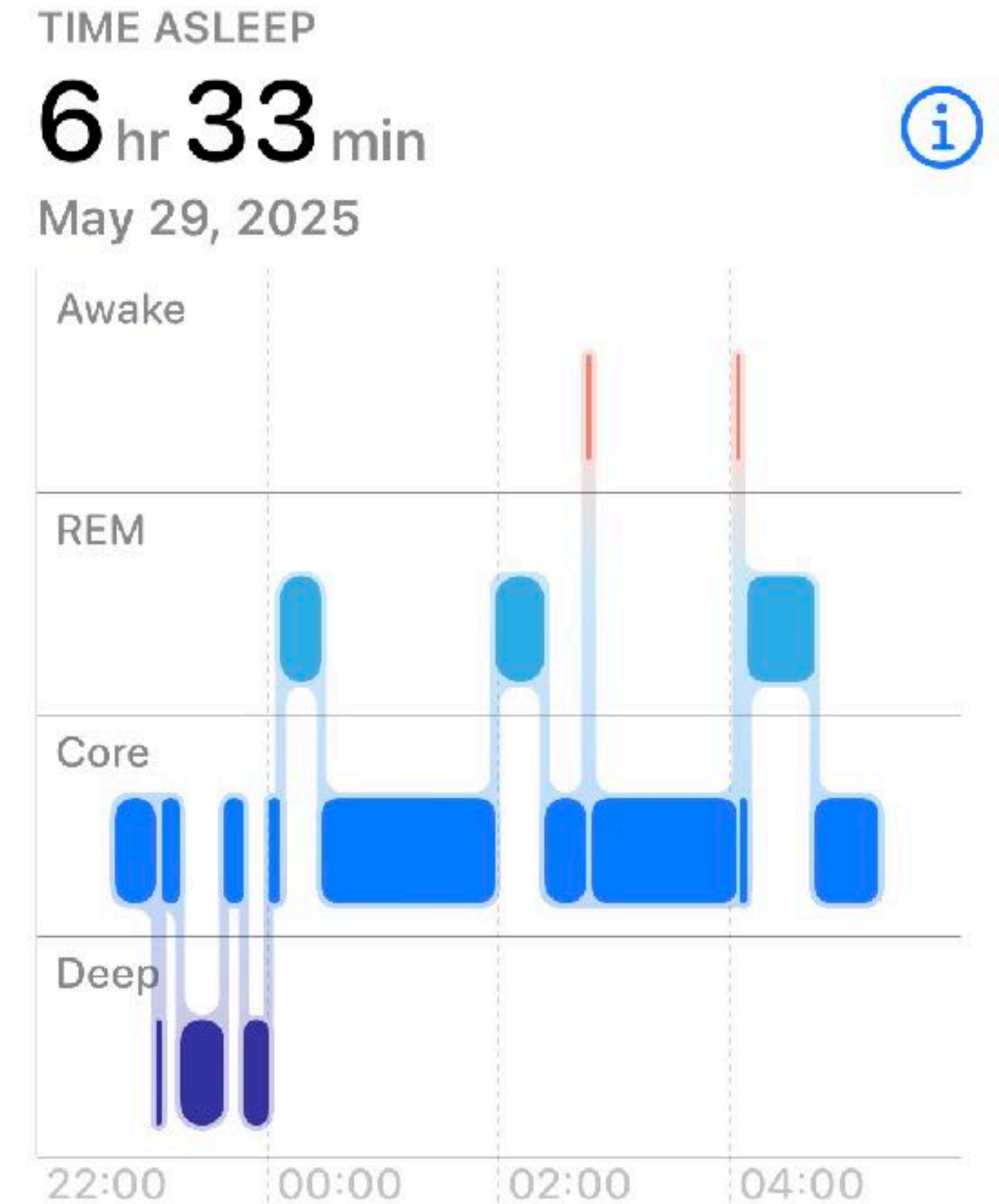
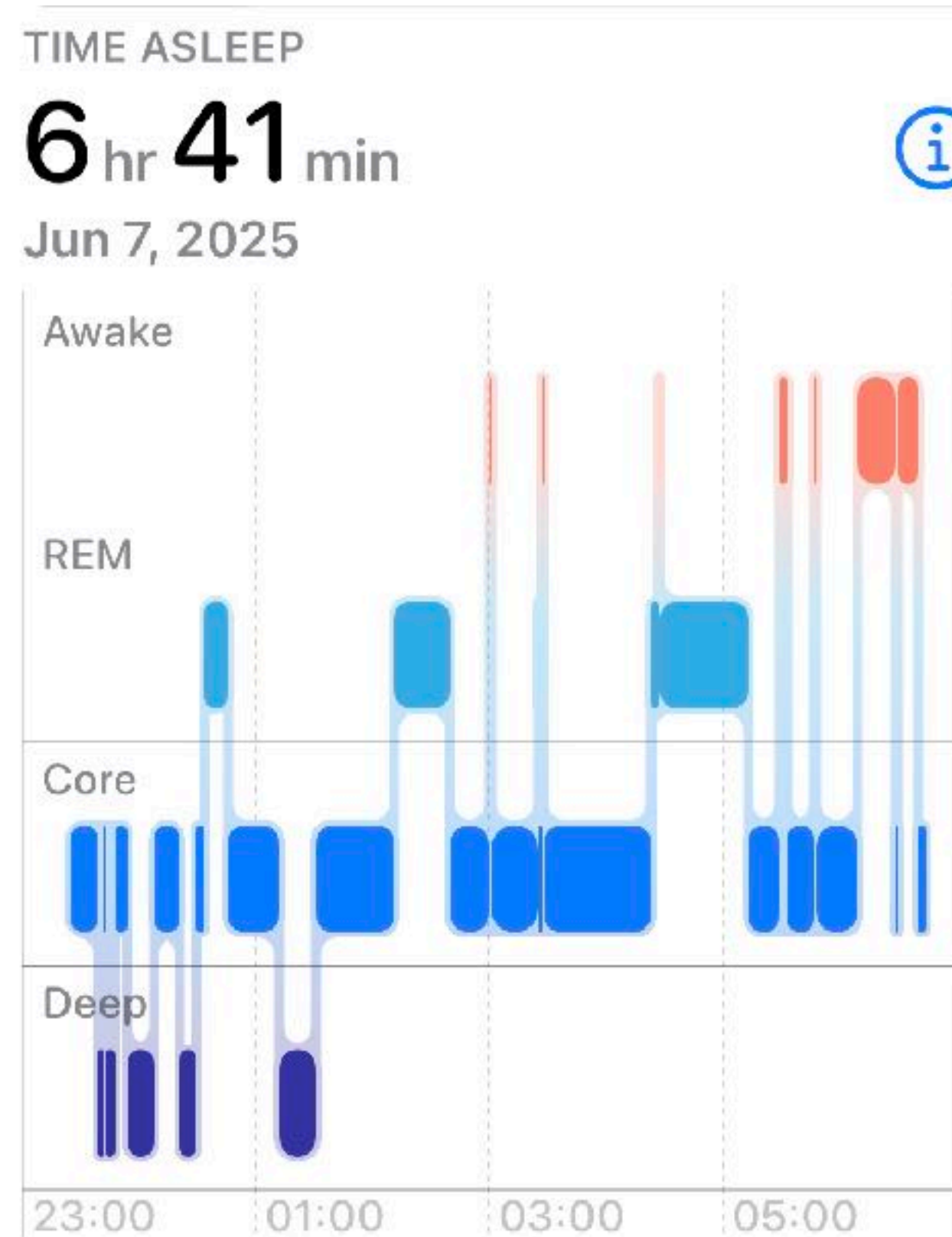
Morning super soldier

- According to a handful of studies carried out by the NIH, Caffeine is at its peak efficacy when consumed with a 2:1 ratio of L-Theanine.
- This is because L-Theanine counters the “downing” effect of caffeine where it causes jitters/anxiety etc
- Excerpt:
When used in isolation, both have a proven track record of success. When used in combination, similar outcomes are observed but the necessary effective dosage may be reduced considerably, in particular for caffeine. With this knowledge, those who desire the effects of caffeine but who may be sensitive to higher dosages may consider the combination of the two ingredients, while utilizing a lower dosage of each (e.g., 50-100 mg caffeine; 100-200 mg L-theanine). This combination may be helpful for improvements in certain aspects of cognitive performance and mood.

Melatonin

Helps you stay asleep

- Important to get as much REM and Deep sleep possible, since that is how our body truly repairs itself.
- If you wake up often at night, melatonin will alleviate that, but it will not help you fall asleep in the first place



END

